

BOSTON LOGAN, MA, USAWMO: **725090**Lat: **42.361N**Lon: **71.010W**Elev: **12**StdP: **14.69**Time Zone: **-5.00 (NAE)**Period: **94-19**WBAN: **14739****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.7	12.8	-10.8	3.1	10.6	-6.3	3.9	15.9	31.8	36.6	28.3	31.0	16.0	300	0.602

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.7	90.8	73.0	87.7	71.6	84.5	70.0	76.0	86.1	74.4	83.2	72.9	80.7	13.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.8	121.8	80.8	71.4	116.1	79.0	70.2	111.0	77.8	39.3	86.4	37.8	83.2	36.4	80.8	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.7	23.9	20.6	DB	2.2	95.7	5.0	3.0	-1.4	97.8	-4.3	99.6	-7.1	101.2	-10.7	103.4	(4)
(5)			WB	0.3	78.4	4.5	1.5	-3.0	79.5	-5.6	80.4	-8.2	81.2	-11.4	82.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.2	29.9	32.0	38.4	48.8	58.3	67.9	74.3	72.8	66.0	55.2	45.0	36.1	(6)	
	DBStd	17.16	10.16	9.04	8.65	7.78	7.59	7.36	5.81	5.48	6.35	7.14	8.05	8.60	(7)	
	HDD50	2284	625	507	373	111	9	0	0	0	0	30	192	437	(8)	
	HDD65	5498	1088	925	824	490	238	53	3	4	62	314	602	895	(9)	
	CDD50	3072	2	2	15	75	267	536	752	707	479	190	40	7	(10)	
	CDD65	812	0	0	1	4	31	139	290	246	91	10	0	0	(11)	
	CDH74	5864	0	0	8	59	280	1005	2290	1672	495	54	1	0	(12)	
	CDH80	1920	0	0	2	19	89	330	822	519	132	7	0	0	(13)	
(14) Wind	WSAvg	11.2	12.4	12.3	12.6	12.0	10.8	10.1	9.8	9.5	10.1	11.1	11.2	11.9	(14)	
(15) Precipitation	PrecAvg	42.4	3.5	3.4	3.9	3.5	3.3	3.5	3.0	3.2	3.4	3.6	3.9	4.1	(15)	
	PrecMax	53.6	10.6	8.0	14.9	9.6	12.5	13.2	7.6	8.0	9.8	9.6	9.3	9.7	(16)	
	PrecMin	23.7	0.6	0.7	0.6	0.8	0.5	0.0	0.6	0.7	0.7	0.4	0.6	0.8	(17)	
	PrecStd	6.7	1.9	1.7	2.3	1.9	2.0	2.7	1.6	1.9	2.0	2.1	2.0	2.0	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.1	60.8	71.4	82.4	88.2	92.4	95.1	93.6	89.9	80.2	69.4	63.4	(19)	
		MCWB	55.8	54.1	56.1	62.4	67.8	73.5	76.0	75.0	73.0	65.6	61.3	57.5	(20)	
	2%	DB	53.9	53.4	61.2	71.7	81.7	87.9	91.0	89.1	83.5	73.9	64.4	56.8	(21)	
		MCWB	50.0	47.2	51.8	57.4	64.8	71.3	73.4	73.0	69.9	64.4	58.1	52.2	(22)	
	5%	DB	48.0	48.4	55.1	65.6	75.7	83.6	88.0	85.5	79.6	69.9	60.6	52.5	(23)	
		MCWB	43.6	42.5	47.0	54.2	62.0	68.9	72.2	71.2	68.0	62.1	54.2	47.7	(24)	
	10%	DB	43.2	44.1	50.2	60.6	70.6	79.9	84.7	82.3	75.8	65.8	56.8	48.3	(25)	
		MCWB	39.0	38.6	43.1	51.2	59.6	67.2	70.7	70.0	66.5	58.4	51.1	43.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.3	56.0	59.3	64.5	70.7	76.3	78.2	77.7	75.1	70.2	63.8	58.6	(27)	
		MCDB	59.0	59.3	67.7	76.1	84.0	88.0	90.4	89.5	84.2	75.0	67.9	61.5	(28)	
	2%	WB	50.8	48.5	53.7	60.2	67.7	73.8	76.3	75.6	72.5	66.5	59.8	53.3	(29)	
		MCDB	52.9	52.3	59.8	69.2	77.8	84.3	86.7	84.5	79.3	71.8	63.5	56.4	(30)	
	5%	WB	44.3	43.3	48.2	56.6	64.4	71.7	74.6	74.1	70.8	63.3	55.5	48.8	(31)	
		MCDB	47.3	47.0	53.5	63.0	72.5	79.9	83.7	81.6	76.7	68.4	59.2	51.7	(32)	
	10%	WB	39.6	39.5	43.8	52.2	61.5	69.5	73.0	72.5	68.8	60.1	52.3	44.4	(33)	
		MCDB	42.5	43.1	48.8	59.2	69.2	76.4	81.1	79.4	74.4	64.5	55.7	47.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.9	13.4	13.4	14.6	14.8	14.8	14.7	14.0	13.7	13.5	12.6	12.0	(35)	
		MCDBR	18.4	18.5	21.5	23.8	23.9	20.5	19.2	18.6	18.8	18.5	17.2	16.7	(36)	
	5% WB	MCWBR	17.0	14.7	14.7	13.5	11.8	9.9	7.9	7.9	9.3	11.7	13.5	15.2	(37)	
		MCWBR	18.7	17.3	20.8	21.9	21.0	19.3	17.1	16.5	16.0	17.1	15.5	16.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.293	0.302	0.325	0.370	0.406	0.452	0.472	0.434	0.376	0.335	0.311	0.295	(40)	
		taud	2.505	2.493	2.487	2.392	2.329	2.234	2.185	2.278	2.423	2.543	2.571	2.568	(41)	
	Ebn at Noon		273	288	293	285	276	262	255	262	272	272	265	263	(42)	
		Edh at Noon	23	27	30	36	40	44	45	40	32	25	21	20	(43)	
(44) All-Sky Solar Radiation	RadAvg		559	834	1153	1482	1650	1783	1878	1689	1357	901	611	462	(44)	
	RadStd		36	71	80	159	183	168	114	75	92	78	48	50	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+169	+86
(47) Regional (72 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+166	+76

Nomenclature: See separate page